

ROMAN KUDERMIN

Mathematics & ML Engineer

✉ romankudermine@gmail.com

in roman-kudermine | github.com/bebz0 | 📍 Kyiv, Ukraine

SUMMARY

Second-year Computer Mathematics student with hands-on experience in NLP, LLMs. Passionate about bridging theoretical mathematics with practical deep learning, demonstrated through academic research, intensive NLP training, and award-winning hackathon performances. Seeking a junior ML/NLP role to apply my strong mathematical foundation to data-driven solutions.

EDUCATION

Taras Shevchenko National University of Kyiv (KNU)

Kyiv, Ukraine

Bachelor of Science in Computer Mathematics (Mechanics and Mathematics Faculty)

Sep. 2024 – Present

- Specialization: Mathematics, Algorithms, and Computer Science
- Relevant Coursework: Algorithms & Data Structures, Mathematical Analysis, Linear Algebra, Discrete Mathematics, Probability Theory, Algebra & Number Theory.
- Academic Rating: **94.6 / 100**

EXPERIENCE

Confidential Company (NDA)

Remote

ML/NLP Research Assistant

Jan. 2026 – Mar. 2026

- Assisted in developing NLP pipelines for text classification and named entity recognition tasks.
- Preprocessed and annotated text datasets to prepare training data for transformer-based models.
- Stack: Python, HuggingFace Transformers, NumPy, Git.

TECHNICAL SKILLS

Programming: Python, C/C++, SQL (PostgreSQL, MySQL), Octave.

Deep Learning & NLP: PyTorch, HuggingFace Transformers, LLM Distillation, RAG.

Data Science & ML: Scikit-learn, XGBoost, Pandas, NumPy, SciPy, Matplotlib, Seaborn.

Backend & Deployment: FastAPI, Django, Docker, MLflow.

Tools & OS: Git/GitHub, Linux, LaTeX.

Languages: Ukrainian (Native), English (B2), French (A2).

HACKATHONS & PROJECTS

KSE NLP Summer School: From LLM to Agents & Retrieval | KSE & Superhuman

June 2026

- Participated in an intensive 4-day advanced NLP program focused on LLM architectures beyond Transformers, retrieval systems, and agentic workflows.
- Studied modern techniques for LLM distillation, fine-tuning, tokenizer adaptation (LapaLLM-style), synthetic data generation, and AI Safety & Interpretability.
- Gained practical insights into MLOps for LLMs, agentic harness frameworks, Kaggle NLP winning strategies, and OCR model training (Rukopys).

Monte Carlo-Based Ablation Study of CNNs | PyTorch, Python

2026

- Engineered a custom PyTorch pipeline to evaluate Monte Carlo Dropout for estimating predictive uncertainty in Convolutional Neural Networks: VGG16 vs ResNet18.
- Designed and executed ablation experiments to compare deterministic inference and stochastic evaluation strategies.
- Demonstrated how stochastic sampling improves model calibration and provides robust confidence intervals for deep learning predictions.

ML Forecasting Hackathon | N-BEATS, PyTorch, Python

2025

- Secured **2nd Place** out of 12 teams in a time-series forecasting challenge.
- Implemented N-BEATS architecture with custom seasonality decomposition.
- Achieved MAPE of 16% on the test set, outperforming the provided baseline model.

Research: Complex Numbers in Algebra | Junior Academy of Sciences (MAN)

2024

- Conducted independent research on applications of complex numbers in algebraic structures.
- Defended the research paper at the regional competition; work reviewed by a university-level jury.

1st Place ($\times 2$), University Mathematical Battles

2025

- Achieved 1st place in both rounds of a university-level mathematics olympiad.